



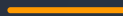
Gennoor Tech

TRAIN. INNOVATE. BUILD.

5-DAY TECHNICAL BOOTCAMP · 40 HOURS

AI on AWS

Amazon Bedrock, SageMaker, Lex, Textract, and the complete AWS AI stack



5

DAYS

40

HOURS

20+

HANDS-ON LABS

6

AWS PROJECTS

Course Overview

This 5-day hands-on bootcamp covers the full AWS AI and ML platform — from Amazon Bedrock and SageMaker through Lex, Comprehend, Textract, Rekognition, and serverless AI patterns. Developers build 6 production-ready AI solutions using AWS's AI stack, learning to leverage its strengths in model choice, serverless architecture, and global infrastructure.

AWS offers the widest choice of AI models and the deepest serverless integration. Amazon Bedrock gives you access to Claude, Llama, Mistral, Titan, and more through a single API. SageMaker provides the most mature ML platform for custom training and deployment. This bootcamp covers the entire ecosystem — from managed AI services for rapid development to custom ML pipelines for full control.

Who Is This For?



AWS Developers

Engineers building AI solutions on the AWS stack



Cloud Architects

Architects designing AI solutions with AWS services



ML Engineers

Engineers deploying models on SageMaker



Multi-Cloud Teams

Teams evaluating AWS AI alongside Azure/GCP

Learning Outcomes

- 1 **Build with Amazon Bedrock** — access Claude, Llama, Mistral, and Titan through a unified API with guardrails, knowledge bases, and agents
- 2 **Master Amazon SageMaker** — custom model training, hyperparameter tuning, model registry, real-time and batch inference endpoints
- 3 **Implement RAG on AWS** — Bedrock Knowledge Bases with S3, OpenSearch Serverless vector search, and retrieval-augmented generation
- 4 **Build conversational AI** — Amazon Lex bots with Lambda fulfillment, multi-turn flows, and integration with Connect for voice
- 5 **Process documents and media** — Amazon Textract for document extraction, Comprehend for NLP, Rekognition for image/video analysis
- 6 **Deploy serverless AI** — Lambda + API Gateway for inference, Step Functions for ML workflows, and EventBridge for event-driven AI

Prerequisites

Python programming (intermediate), basic AWS familiarity (console, IAM, S3). Experience with REST APIs. An AWS account with Bedrock model access is recommended (free tier works for most labs).

5-Day Curriculum

Entirely hands-on with AWS console and SDK labs. Schedule and session order are fully customizable based on your organization's needs and time zones.

01

Amazon Bedrock — Generative AI on AWS

LIVE CODING + HANDS-ON LABS

SESSION A

Bedrock Fundamentals

- AWS AI ecosystem overview: Bedrock, SageMaker, AI Services
- Bedrock model access: Claude, Llama, Mistral, Titan, Cohere
- Converse API: unified interface across all models
- Lab: Build a multi-model chat application with Bedrock
- Bedrock Guardrails: content filtering, topic denial, PII redaction

SESSION B

Bedrock Agents & Knowledge Bases

- Bedrock Knowledge Bases: S3 data sources, chunking, embeddings
- OpenSearch Serverless for vector search
- Bedrock Agents: action groups, function calling, orchestration
- Lab: Build a RAG pipeline with Bedrock Knowledge Bases
- Lab: Build a Bedrock Agent that queries databases and APIs

02

Amazon SageMaker — Custom ML at Scale

HANDS-ON LABS + ARCHITECTURE WORKSHOP

SESSION A

SageMaker Core Platform

- SageMaker Studio: notebooks, experiments, feature store
- Built-in algorithms: XGBoost, linear learner, image classification
- Custom training: bring your own container (BYOC)
- Lab: Train and deploy a fraud detection model on SageMaker
- Lab: Hyperparameter tuning with SageMaker Automatic Model Tuning

SESSION B

SageMaker Pipelines & MLOps

- SageMaker Pipelines: automated ML workflows
- Model Registry: versioning, approval, deployment promotion
- SageMaker Model Monitor: data drift, model quality
- Lab: Build an end-to-end ML pipeline with SageMaker Pipelines
- Lab: Set up model monitoring with automated retraining

03

NLP, Conversational AI & Document Processing

HANDS-ON LABS + INDUSTRY PROJECTS

SESSION A**Amazon Lex & Comprehend**

- Amazon Lex V2: intents, slots, fulfillment with Lambda
- Multi-turn conversation design and context management
- Amazon Comprehend: sentiment, entities, key phrases, PII detection
- Lab: Build a customer service chatbot with Lex + Lambda
- Lab: Build a document sentiment analysis pipeline with Comprehend

SESSION B**Amazon Textract & Translate**

- Textract: tables, forms, key-value pairs, expense analysis
- Textract Queries: ask specific questions about documents
- Amazon Translate: real-time and batch translation
- Lab: Build an invoice processing system with Textract
- Lab: Build a multilingual document pipeline (Textract + Translate + Comprehend)

04

Computer Vision, Speech & Serverless AI

HANDS-ON LABS + SERVERLESS WORKSHOP

SESSION A**Amazon Rekognition & Transcribe**

- Rekognition: face detection, label detection, content moderation
- Rekognition Custom Labels: train custom image classifiers
- Amazon Transcribe: speech-to-text, speaker diarization
- Amazon Polly: text-to-speech with neural voices
- Lab: Build a content moderation pipeline for user-uploaded images

SESSION B**Serverless AI Patterns**

- Lambda for inference: cold starts, layers, container images
- API Gateway + Lambda for AI APIs
- Step Functions for multi-step ML workflows
- EventBridge for event-driven AI processing
- Lab: Build a serverless document processing pipeline (S3 → Lambda → Textract → DynamoDB)

05

Advanced Bedrock, Security & Capstone

CAPSTONE PROJECT + ARCHITECTURE REVIEW + PRESENTATIONS

SESSION A**Advanced Bedrock & Fine-Tuning**

- Bedrock model customization: fine-tuning and continued pre-training
- Bedrock Flows: visual workflow builder for AI applications
- PartyRock: no-code AI app builder on Bedrock
- Cost optimization: on-demand vs provisioned throughput
- Lab: Fine-tune a Titan model on custom enterprise data

SESSION B**Capstone: Enterprise AWS AI Solution**

- Team project: End-to-end AWS AI solution
- Bedrock + SageMaker + Textract + Lambda integration
- Architecture review: IAM, VPC, encryption, cost, compliance
- Team presentations and expert feedback
- AWS AI certification paths and next steps

Flexible & Customizable: Session order, timing, depth, and industry focus can be tailored to your organization's specific requirements. Content is regularly updated to reflect the latest AWS AI developments.

AWS AI Services Covered

All services used hands-on with AWS console and SDK labs

- Amazon Bedrock
- Amazon SageMaker
- Amazon Lex V2
- Amazon Comprehend
- Amazon Textract
- Amazon Rekognition
- Amazon Transcribe
- Amazon Polly
- Amazon Translate
- AWS Lambda
- API Gateway
- Step Functions
- OpenSearch Serverless
- S3
- DynamoDB

What You Take Home

**6 AWS AI Projects**
Complete code — Bedrock chat app, RAG pipeline, Lex bot, Textract extractor, SageMaker model, serverless AI pipeline

**AWS AI Architecture Guide**
Reference patterns for enterprise AWS AI: IAM policies, VPC endpoints, encryption, multi-account strategies, cost optimization

**CloudFormation Templates**
Infrastructure-as-Code templates for Bedrock, SageMaker endpoints, Lambda layers, and serverless AI pipelines

**AWS vs Azure vs GCP Matrix**
Detailed comparison of AI services across all three clouds — features, pricing, model access, migration patterns

Certification of Completion

**Gennoor Tech — Certificate of Completion**
Upon completing all 5 days (40 hours), each participant receives a **Gennoor Tech Certificate of Completion** in "AI on AWS". Digitally signed by the Founder & Director. Validates proficiency in Amazon Bedrock, SageMaker, Lex, Textract, Rekognition, and enterprise AWS AI architecture.

Recommended AWS Certifications

AWS MLS-C01
Machine Learning Specialty

AWS SAA-C03
Solutions Architect Associate

AWS DEA-C01
Data Engineer Associate

AWS AIF-C01
AI Practitioner

Designed & Delivered By

Jalal Ahmed Khan
Founder & Director, Gennoor Tech Private Limited
Senior AI Consultant and **Microsoft Certified Trainer (MCT)** with 14+ years of experience delivering AI programs across Azure, AWS, and GCP. Multi-cloud expertise covering Bedrock, SageMaker, Azure AI, and Vertex AI. Delivered AI training for Microsoft, EY, IBM, Boeing, and 80+ organizations across 6+ countries.
12+ active certifications across Microsoft, GitHub, and cloud platforms. Production experience with serverless AI pipelines and enterprise document processing systems. **M.E. Mechanical Engineering**, Mumbai University.

Delivery Options

 **On-Site** — customized at your office |  **Virtual Live** — interactive via Teams/Zoom |  **International** — GCC, Africa, APAC, UK



Ready to Build AI on AWS?

Contact us to discuss a customized bootcamp
for your organization or register for the next open batch.

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